

Narrative Hook

The school woodshop just got a donation of lumber, and the carpentry teacher needs help — fast. She's trying to build a right-angled corner brace for a bookshelf, but her only measuring tools are a tape measure and a pencil. Can your math class save the day by proving that a triangle with sides 3, 4, and 5 units always makes a perfect right angle?

Timeline

PHASE	ACTIVITY	TIME
1	Narrative hook & prior knowledge activation	5 min
2	Direct instruction — Pythagorean theorem	10 min
3	Hands-on woodshop measurement activity	20 min
4	Mathematical verification & calculations	10 min
5	Group presentations & debrief	10 min
6	Reflection journaling	5 min
Total	60 min	

>  **[TEACHER: Pacing note]** This activity is designed for a 60-minute single period, but Phases 3–5 are ambitious for that window — particularly if material distribution, gluing/taping, and full-group presentations all occur in one sitting. **Strongly recommended: split across two periods.** Period 1 = Phases 1–4 (hook through calculation); Period 2 = Phase 5 presentations + Phase 6 reflection. If teaching in a single period, consider limiting presentations to 2–3 representative groups (see Step 9) and using pre-cut, pre-labelled strips to reduce Phase 3 setup time.

Materials

- Pre-cut scrap wood strips (or heavy cardboard as a low-cost substitute) — lengths of 30 cm, 40 cm, and 50 cm per group
- Tape measures or rulers (one per group)
- Pencils and graph paper
- A carpenter's square or set square (for verification)
- String or thin rope (optional, for the "rope-stretcher" historical demonstration)
- Glue or masking tape (to temporarily join triangle corners)
- Worksheets with blank Pythagorean equation templates ($a^2 + b^2 = c^2$)

Step-by-Step Instructions

- 1. Open with the hook.** Read the bookshelf scenario aloud. Ask students: *"How would you check if a corner is perfectly square without a protractor?"* Record student ideas on the board.
- 1. Direct instruction (10 min).** Introduce the Pythagorean theorem: $a^2 + b^2 = c^2$. Show how it applies to right triangles. Draw a labeled diagram on the board. Emphasize that this is the exact method carpenters and builders use in real life.
- 1. Form groups.** Divide students into groups of 3–4. Distribute the pre-cut wood strips (or cardboard) in the 3:4:5 ratio (30 cm, 40 cm, 50 cm). *(Tip: pre-label strips before class to keep Phase 3 within its 20-minute window — see Timeline pacing note.)*
- 1. Build the triangle.** Each group joins their three strips at the ends using tape or glue to form a triangle. Lay it flat on the desk.
- 1. Predict before measuring.** Ask each group to predict which angle is the right angle and mark it with a pencil dot. Record their prediction on the worksheet.
- 1. Test the corner.** Students place their carpenter's square (or fold a piece of paper to make a right angle) at the marked corner to check whether it is truly 90°. Record the result.

1. **Verify with math.** Students plug their measured side lengths into $a^2 + b^2 = c^2$ on their worksheet and confirm the equation holds. Discuss: *What does it mean that the math and the physical test agree?*
 1. **Real-world extension.** Pose a new scenario: *"A doorframe in the woodshop is 90 cm wide and 120 cm tall. What length should the diagonal brace be?"* Groups solve on graph paper and present their answer. *(If time is tight in a single-period setting, this problem can be assigned as homework or completed at the start of Period 2 — see Timeline pacing note.)*
 1. **Group presentations (10 min).** Each group shares their triangle, their measurements, and their diagonal brace answer. Encourage peer questions.
- > **Opt-out / alternative:** Any student who is not comfortable presenting aloud to the full class may instead submit a written summary (on their worksheet or graph paper) covering the same three elements — triangle construction, measurements, and brace calculation — to be reviewed by the teacher individually. This written submission is assessed on the same rubric criteria. *If running a single period, consider selecting 2–3 groups to present aloud and having remaining groups submit written summaries, which also helps keep Phase 5 within 10 minutes.*
1. **Reflection journaling (5 min).** Students respond to the reflection questions independently in writing.

Reflection Questions

1. Why do carpenters and builders need to understand right angles? What could go wrong if a corner is even slightly off?
2. In your own words, explain what the Pythagorean theorem tells us about the sides of a right triangle.
3. The 3-4-5 method has been used by builders for thousands of years — long before formal algebra existed. What does that tell you about the relationship between math and practical trades?
4. If you only knew two sides of a right triangle, how would you find the third? Walk through the steps.
5. Where else in everyday life — beyond woodworking — might you need to calculate a missing length or check a right angle?

 Assessment Rubric

CRITERION	DEVELOPING (1)	MEETS EXPECTATIONS (2)	EXCEEDS EXPECTATIONS (3)
Conceptual Understanding	Cannot explain what $a^2 + b^2 = c^2$ means — <i>e.g., leaves the definition blank, or says only "it's a formula" with no reference to sides or triangles</i>	Correctly states the theorem and identifies sides — <i>e.g., writes "the two shorter sides squared and added equal the longest side squared" and correctly labels a, b, c on a diagram</i>	Explains why the theorem works and connects it to the physical triangle — <i>e.g., describes how the physical test with the carpenter's square confirmed the mathematical equation, or explains the role of the hypotenuse</i>
Accurate Calculation	Major errors in arithmetic or formula setup — <i>e.g., adds sides without squaring ($3+4=5$), or substitutes values into the wrong positions in the formula</i>	Correct formula setup with minor arithmetic errors — <i>e.g., correctly writes $30^2 + 40^2 = c^2$ and squares the values but makes a single arithmetic slip in the addition or square root step</i>	All calculations correct; diagonal brace problem solved accurately — <i>e.g., correctly computes $90^2 + 120^2 = 22500$ and states the diagonal brace is 150 cm</i>
Hands-On Application	Triangle not correctly assembled or angle not tested — <i>e.g., strips are not joined at their endpoints, or the carpenter's square is applied to the wrong corner</i>	Triangle assembled and angle tested with minor procedural gaps — <i>e.g., triangle is correctly built and tested but the result is not recorded on the worksheet</i>	Triangle built precisely; right angle verified using both physical and mathematical methods — <i>e.g., carpenter's square confirms 90° at the predicted corner AND worksheet shows $a^2 + b^2 = c^2$ confirmed with measured values</i>

Communication & Presentation	Explanation is unclear or incomplete — <i>e.g., presents only the final answer with no description of process, or written summary omits two or more required elements (triangle construction, measurements, brace calculation)</i>	Explains findings to group with basic clarity — <i>e.g., states what the triangle sides measured and whether the right angle was confirmed, either in a verbal presentation or written summary</i>	Clearly presents process, results, and real-world connection; responds to peer questions (or addresses likely questions in written summary) — <i>e.g., explains how the 3-4-5 ratio produces a right angle, connects it to the bookshelf scenario, and either answers a peer question or anticipates one in writing</i>
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Note on Communication criterion: "presentation" includes both a verbal group presentation and the written summary alternative described in Step 9. Both formats are assessed on the same indicators above.

Customization Zones

- 1. Adjust the real-world scenario** to match a project your woodshop teacher is actually working on — even a simple picture frame or garden box works perfectly.

 [INSERT TEACHER NOTE HERE]

- 1. Swap materials based on budget** — if wood is unavailable, drinking straws, popsicle sticks, or strips of paper taped together replicate the triangle activity at near-zero cost.

 [INSERT PHOTO OR IMAGE HERE]

- 1. Scaffold the worksheet** for students who need additional support by pre-filling the formula template and providing a worked example alongside the blank problems.

 [INSERT STUDENT WORKSHEET HERE]

- 1. Extend for advanced students** by challenging them to find other Pythagorean triples (e.g., 5-12-13, 8-15-17) and test whether those also produce right angles with physical materials.

 [INSERT TEACHER NOTE HERE]